

Linear Algebra

COSMOS Summer School
University of California, Santa Cruz

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Summer 2025

Remark 0.1. The following problem sets are compiled from a linear algebra course taught in Summer 2025. Max Su, a fellow graduate student and TA for the course helped write the project on Linear Regression (admittedly the most well written of the bunch). None of the problems were explicitly taken from other books, but as it is a fairly standard course I'm sure some appear elsewhere. There are also a few errors throughout, so beware!

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1 Worksheet 1: Linear Equations

Problem 1. Draw a picture of the solution set for each of the following equations in $\mathbb{R}^2$,

1. $x = 0$

2. $y = 3$

3. $y - x = 0$

4. $4x - 2y = 1$

Problem 2. Show that the equation,

$$32x + 20y = 7$$

has no integer solutions. What is the smallest positive integer you can write in the form, $32x + 20y$?

Problem 3. Show that the equation

$$2x + 17y = 1$$

has an integer solution. Then show that every integer can be written in the form,

$$2z + 17t$$

What values of z, t give $2z + 17t = 100$?

Problem 4. Show that the following equations,

$$24x + 8y = 32$$

$$6x + 2y = 8$$

Have the same set of solutions.

Problem 5. Show that there is no pair $(x, y) \in \mathbb{R}^2$ that satisfies the following equations simultaneously,

$$2x + 3y = 9$$

$$4x + 6y = 17$$

Problem 6. Show that there is a unique pair $(x, y) \in \mathbb{R}^2$ that satisfies the following equations,

$$2x + 3y = 1$$

$$7x + 9y = 13$$

Problem 7. Use geometry to argue that a general pair of equations

$$ax + by = v$$

$$cx + dy = w$$

Either has no solutions, one solution, or infinitely many solutions. Hint: Each equation cuts out a line in the plane, a solution to both must lie in the overlap. Draw a picture of each case.

Problem 8. Suppose not all of a, b, c, d are zero. Show that the pair of equations,

$$ax + by = v$$

$$cx + dy = w$$

has infinitely many solutions if and only if,

$$ad - bc = 0$$

$$dv - bw = 0$$

Hint: Write both equations in slope intercept form. There are infinitely many solutions if and only if they are the *same* line.

2 Worksheet 2: Matrices

Question 2.1: Use row-reduction to determine whether or not the following equations have solutions. In each case determine whether there are no solutions, one solution, or infinitely many solutions.

$$\text{i) } \begin{bmatrix} 3 & 21 \\ 17 & 8 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \text{ii) } \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad \text{iii) } \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 17 \\ 34 \end{bmatrix}$$

Question 2.2: Compute the following products.

$$\text{i) } \begin{bmatrix} 3 & 21 \\ 17 & 8 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{ii) } \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \text{iii) } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{iv) } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

In each case try to guess what the effect of repeatedly applying the matrix n times will be. In other words, what is the output of $M^n \cdot v$?

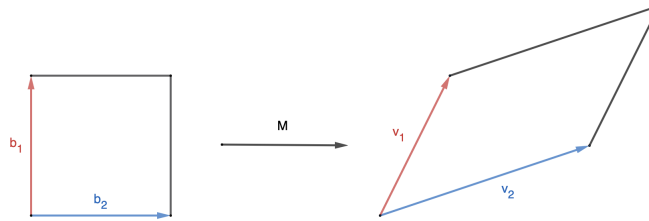
Question 2.3: *Matrices as linear transformations*

We can see from the formula that,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}$$

So a matrix is determined by where it sends these two vectors, $b_1 = [1, 0]$ and $b_2 = [0, 1]$.



Draw pictures of the effects of the following matrices on the unit square. In each case, label where the vectors b_1, b_2 are sent,

$$\text{i) } \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \text{ii) } \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}, \quad \text{iii) } \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \text{iv) } \begin{bmatrix} -3 & 2 \\ 6 & -4 \end{bmatrix}$$

Now try to go backwards. Write down matrices which have the following effects,

1. Swap b_1 and b_2
2. Rotate b_1, b_2 by 90 degrees then shift both three units to the right.
3. Shift both b_1, b_2 three units to the right then rotate by 90 degrees.
4. Rotate b_1, b_2 by 90 degrees, then send b_1 to 3 times b_2 .

Question 2.4: The purpose of this problem is to give a geometric meaning to the number,

$$\det(M) = ad - bc$$

which is called the **determinant** of the matrix,

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We've seen above that the two column vectors $v_1 = (a, c)$ and $v_2 = (b, d)$ bound two sides of a parallelogram in the (x, y) -plane.

- (a) Draw a picture and use the Pythagorean theorem to convince yourself that the length of v_1 is given by,

$$||v_1||^2 = a^2 + c^2$$

and similarly,

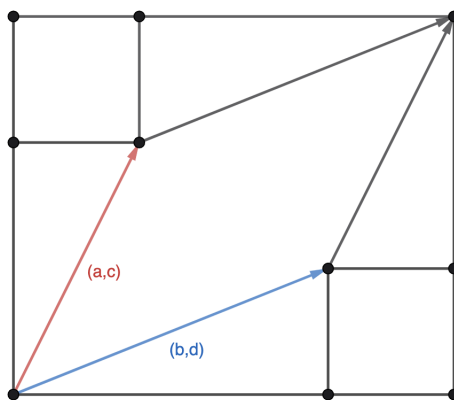
$$||v_2||^2 = b^2 + d^2$$

- (b) Use the formula Area=base · height for the area of a parallelogram to convince yourself that the area of the parallelogram above is given by,

$$A = ||v_1|| \cdot ||v_2|| \sin \theta$$

where θ is the angle between the vectors.

- (c) Bound this parallelogram in a larger rectangle as follows,



Compute the area of the total rectangle and subtract off the outer areas to see that the area of the parallelogram is,

$$A = |ad - bc|$$

So the determinant is equal to this area!

$$ad - bc = \|v_1\| \cdot \|v_2\| \sin(\theta)$$

For what values of θ is the determinant equal to zero? What does this mean geometrically for the vectors v_1, v_2 ?

(d) Use the trigonometric identity,

$$\sin(\theta)^2 + \cos(\theta)^2 = 1$$

to write,

$$\|v_1\|^2 \cdot \|v_2\|^2 = \|v_1\|^2 \cdot \|v_2\|^2 \sin(\theta)^2 + \|v_1\|^2 \cdot \|v_2\|^2 \cos(\theta)^2$$

Substitute what he have learned from the previous sections to see that,

$$(a^2 + c^2)(b^2 + d^2) = (ad - bc)^2 + X^2$$

where $X = \|v_1\| \cdot \|v_2\| \cos(\theta)$ is the unknown quantity we would like to solve for. Solve and show that,

$$\|v_1\| \cdot \|v_2\| \cos(\theta) = ab + cd$$

This is known as the **dot product** of the vectors v_1, v_2 ,

$$v_1 \cdot v_2 = \begin{bmatrix} a & c \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = ab + cd = \|v_1\| \cdot \|v_2\| \cos(\theta)$$

When does $v_1 \cdot v_2 = 0$? What does this mean geometrically for the vectors v_1, v_2 ?

Question 2.5: In the last problem we saw the dot product,

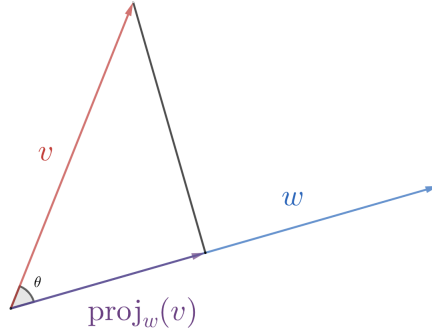
$$v_1 \cdot v_2 = ab + cd = \|v_1\| \cdot \|v_2\| \cos(\theta)$$

where θ is the angle between the vectors $\vec{v}_1, \vec{v}_2$.

(a) Let $\vec{v} = [a, b]$. Show that,

$$\vec{v} \cdot \vec{v} = a^2 + b^2 = \|\vec{v}\|^2$$

(b) Let $\vec{v} = [a, b]$, $\vec{w} = [c, d]$ be vectors. In this problem we consider the following picture.



Show that the vector $\text{proj}_w(v)$ (called the projection of v onto w) is given by the formula,

$$\text{proj}_w(v) = \frac{\vec{v} \cdot \vec{w}}{\|\vec{w}\|^2} \cdot \vec{w}$$

Show that the vector,

$$\vec{v} - \text{proj}_w(v)$$

is orthogonal to w and that this is the vector opposite to the angle θ in the above picture.

(c) Compute the projection of the vector $\vec{x} = [x, y]$ onto the vectors $w_1 = [1, 1]$, $w_2 = [1, -1]$.

Show that,

$$\vec{x} - \text{proj}_{w_1}(x) = \text{proj}_{w_2}(x)$$

Question 2.6: Let,

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be a nonzero matrix. We will be interested in solutions to equations of the form,

$$M \cdot \vec{x} = \vec{v}$$

In other words, what vectors $\vec{v}$ can be written this way?

- (a) First we consider the case where $\det(A) = ad - bc = 0$. Use the formula,

$$\det(A) = ad - bc = \|v_1\| \cdot \|v_2\| \sin(\theta)$$

to show that v_1, v_2 lie on the same line in $\mathbb{R}^2$. Then show that all the vectors w of the form $Av = w$ lie on this same line.

- (b) Suppose $\det(M) = 0$ as in the previous part. Show that there is a nonzero vector $v = [x, y]$ such that,

$$v \cdot v_1 = ax + by = 0$$

Then show that $M \cdot \vec{v} = \vec{0}$.

- (c) Now consider the case where $\det(A) \neq 0$. Show that,

$$\frac{1}{ad - bc} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} d \\ -c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and,

$$\frac{1}{ad - bc} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} -b \\ a \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- (d) *Cramer's Rule* Use the answer from part (c) to show that the solution to the equation,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} v \\ w \end{bmatrix}$$

is given by,

$$x = \frac{\det \begin{bmatrix} v & b \\ w & d \end{bmatrix}}{\det(M)} \text{ and } y = \frac{\det \begin{bmatrix} a & v \\ c & w \end{bmatrix}}{\det(M)}$$

- (e) Suppose $\det(M) \neq 0$. Use your formula from part (c) to show that $Mv = 0$ if and only if $v = 0$.

Composition of linear transformations

Suppose we are given two matrices,

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } N = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

Viewing these as maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ it is natural to try to compose these maps,

$$\mathbb{R}^2 \xrightarrow{M} \mathbb{R}^2 \xrightarrow{N} \mathbb{R}^2$$

to get a new linear transformation.

Question 2.8: Let M, N be matrices as above.

(b) We now introduce a formula for multiplying two matrices,

$$M \cdot N = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Try to guess the result for, $N \cdot M$. Note that here the output is a matrix so we have,

$$\mathbf{Matrix} \cdot \mathbf{Matrix} = \mathbf{Matrix}$$

(c) Show that,

$$(M \cdot N) \cdot v = M \cdot (N \cdot v)$$

$$(N \cdot M) \cdot v = N \cdot (M \cdot v)$$

(d) Give an example of matrices M, N such that,

$$M \cdot N \neq N \cdot M$$

and an example of matrices M, N such that,

$$M \cdot N = N \cdot M$$

Question 2.9: Let λ be a real number. Describe the effect of the following matrix operations in words,

$$\text{i) } \begin{bmatrix} 1 & \lambda \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \text{ii) } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}, \quad \text{iii) } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{iv) } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & \lambda \end{bmatrix}$$

Question 2.10: Show that if $\det(M) \neq 0$ then,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The matrix,

$$M^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

is called the **inverse matrix** of M . The matrix,

$$\text{Id} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

has a special name, it is called the **identity matrix**. Check that,

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

and that,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Question 2.11: Show that if M, N are 2×2 matrices then,

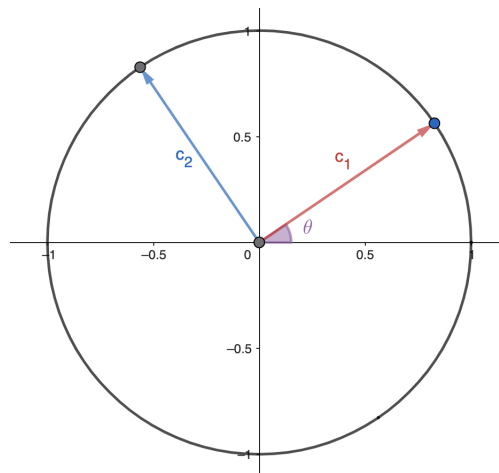
$$\det(M \cdot N) = \det(M) \cdot \det(N)$$

Give an example of two matrices M, N such that,

$$\det(M + N) \neq \det(M) + \det(N)$$

Question 2.12: Rotations: In this we problem construct matrices which describe rotating the plane by some angle θ .

(a) We can visualize the effect of rotating the standard basis vectors in the following picture,



Note that the resulting vectors c_1 and c_2 will again be at a right angle to each other. Work out the (x, y) -coordinates of the vector c_1 , then use the formulas,

$$\sin(\theta + \pi/2) = \cos(\theta)$$

$$\cos(\theta + \pi/2) = -\sin(\theta)$$

to work out the coordinates for c_2 .

(b) Use your answers from part (a) to represent the rotation in matrix form as,

$$R(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

(c) We write $R(\theta)^T$ for the **transpose** of $R(\theta)$. Show that,

$$R(\theta) \cdot R(\theta)^T = R(\theta)^T \cdot R(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(d) Composing two rotations results in a rotation in the sum of the angles,

$$R(\theta) \cdot R(\psi) = R(\theta + \psi)$$

Write this out on both sides to derive formulas for,

$$\sin(\theta + \psi) \text{ and } \cos(\theta + \psi)$$

Then show that,

$$R(\theta)^T = R(-\theta)$$

(e) Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. M is said to be **orthogonal** if $M \cdot M^T = \text{Id} = M^T \cdot M$. Show that M is orthogonal if and only if its column vectors v_1, v_2 have,

$$||v_1||^2 = ||v_2||^2 = 1$$

$$v_1 \cdot v_2 = 0$$

Show that for any matrix $\det(M) = \det(M^T)$. Then conclude that if M is orthogonal,

$$\det(M) = \pm 1$$

3 Worksheet 3: Span, Linear Independence and Basis

Question 3.1: Draw a picture of the span of the following vectors in $\mathbb{R}^3$,

$$\text{i) } \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}, \quad \text{ii) } \text{span}\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}\right\}, \quad \text{iii) } \text{span}\left\{\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}\right\}$$

Question 3.2: Let $\vec{v}_1, \vec{v}_2$ be vectors in $\mathbb{R}^n$. Show that if $\vec{v}_1 = \lambda \vec{v}_2$ for some real number λ , then, $\text{span}\{\vec{v}_1\} = \text{span}\{\vec{v}_2\} = \text{span}\{\vec{v}_1, \vec{v}_2\}$

Question 3.3: Let,

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be a nonzero matrix with column vectors $v_1 = [a, c]$, $v_2 = [b, d]$.

1. Suppose $\det(M) \neq 0$. Use the inverse matrix M^{-1} (or Cramer's rule) to show that every vector $\vec{v}$ can be written,

$$M \cdot \vec{x} = \vec{v}$$

In other words, the image is the whole plane $\text{Im}(M) = \mathbb{R}^2$ and that $\ker(M) = \{\vec{0}\}$.

2. Suppose $\det(M) = ad - bc = 0$. Show that $v_1 = \lambda \cdot v_2$ for some real number λ . Then use the previous exercise to conclude that $\text{Im}(M)$ is a line in $\mathbb{R}^2$.

Use exercise 2.6 from the previous worksheet to show that $\ker(M)$ is a line in $\mathbb{R}^2$.

Question 3.4: Let M be an $n \times n$ matrix, and let $\vec{v}, \vec{w}$ be vectors in $\mathbb{R}^n$. Show that,

$$M \cdot \vec{v} = M \cdot \vec{w}$$

if and only if, $\vec{v} - \vec{w}$ is in $\ker(M)$.

Question 3.5: Informally, a **subspace** $V \subset \mathbb{R}^n$ is a collection of vectors that can be written,

$$V = \text{span}\{v_1, \dots, v_k\}$$

for some finite set of vectors $v_1, \dots, v_k$. A **basis** for V is a collection of vectors $v_1, \dots, v_k$ that span V , but where no proper subset $S \subset \{v_1, \dots, v_k\}$ spans V . The **dimension** of V , written $\dim(V)$ is equal to k where $v_1, \dots, v_k$ is a basis.

1. Show that $\dim(\mathbb{R}^n) = n$. Hint: What is an obvious collection of vectors that spans $\mathbb{R}^n$?

Question 3.6: Use row reduction to compute the image and kernel of the following matrices,

$$\text{i) } \begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 2 \\ 4 & 4 & 12 \end{bmatrix}, \quad \text{ii) } \begin{bmatrix} 3 & 2 & 5 \\ 7 & 3 & 1 \\ 4 & 4 & 2 \end{bmatrix}, \quad \text{iii) } \begin{bmatrix} 2 & 8 \\ 3 & 6 \end{bmatrix}, \quad \text{iv) } \begin{bmatrix} 1 & 18 \\ 3 & 54 \end{bmatrix}$$

In each case, write down a basis for $\ker(M)$ and $\text{Im}(M)$ and show that the dimensions add up to the total dimension,

$$\dim(\mathbb{R}^n) = 2 \text{ or } 3 = \dim(\text{Im}(M)) + \dim(\ker(N))$$

Question 3.7: Here is a more formal definition of a **subspace** $V \subset \mathbb{R}^n$. V is a subspace if,

1. For any two vectors v, w in V , their sum $v + w$ is also in V .
 2. For any vector v in V and any real number λ , the scalar product λv is also in V .
- (a) Let M be an $n \times n$ matrix. Show that $\text{Im}(M)$ and $\ker(M)$ are subspaces in the sense above.
- (b) Show that if $V \subset \mathbb{R}^n$ can be written,

$$V = \text{span}\{v_1, \dots, v_k\}$$

then V is a subspace.

(c) Let $V, W \subset \mathbb{R}^n$ be subspaces. Show that,

$$V + W = \{\vec{v} + \vec{w} \mid \vec{v} \text{ in } V, \vec{w} \text{ in } W\}$$

and,

$$V \cap W = \{\vec{v} \mid \vec{v} \text{ in } V \text{ and } W\}$$

are subspaces.

(d) Show that if,

$$V = \text{span}\{v_1, \dots, v_k\}$$

then there exists a sub-set $B \subseteq \{v_1, \dots, v_k\}$ which is a basis for V .

Question 3.8: The set of solutions to the following equations are **not** subspaces in $\mathbb{R}^2$. In each case, draw a picture of the solutions and explain why it is not a subspace.

(a) $3x - y = 1$

(b) $xy = 0$

(c) $y - x^2 = 0$

(d) $x^2 + y^2 = 1$

Question 3.9: Let,

$$V = \text{span}\{v_1, v_2, \dots, v_k\}$$

be a subspace in $\mathbb{R}^n$. What $n \times n$ -matrix sends all of $\mathbb{R}^n$ onto this subspace?.

Question 3.10: Let $v_1, \dots, v_k$ be a collection of vectors in $\mathbb{R}^n$

(a) We say that the collection $v_1, \dots, v_k$ is **linearly dependent** if there is a combination,

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = \vec{0}$$

where at least *two* of the $\lambda_i \neq 0$. Show that this is the same as saying one of the v_i can be written as a combination of the others.

If no combination of the vectors gives $\vec{0}$ we say the collection $v_1, \dots, v_k$ is **linearly independent**.

- (b) Show that if $\text{span}\{v_1, \dots, v_k\} = \text{span}\{v_1, \dots, v_k, v_{k+1}\}$ then $\{v_1, \dots, v_k, v_{k+1}\}$ is linearly dependent.
- (c) Show that if $B = \{v_1, \dots, v_k\}$ is a basis for a subspace V then for any vector $v \in V$, $\{v, v_1, \dots, v_k\}$ is linearly dependent.
- (d) Let $V, W \subset \mathbb{R}^n$ be subspaces such that $V \subset W$ and $\dim(V) = \dim(W) = k$. Show that $V = W$.

Hint: Assume $V \neq W$. Then there is some $w \in W$ which is not in V . Choose a basis B for V , then show $B \cup \{w\}$ is linearly independent. This means $\dim(W) \geq k + 1$.

- (e) Let M be an $n \times n$ matrix. Show that if the columns form a basis for $\mathbb{R}^n$ then $\text{Im}(M) = \mathbb{R}^n$.
Then show that if $\ker(M) \neq \{\vec{0}\}$, then its columns are linearly dependent.

Question 3.11: Let V be a subspace in $\mathbb{R}^n$ and define,

$$V^\perp = \{\vec{w} \text{ in } V \mid \vec{v} \cdot \vec{w} = 0\}$$

- (a) Let $V = \text{span} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \subset \mathbb{R}^2$ and let $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \subset \mathbb{R}^3$. Draw pictures of $V^\perp$ and $W^\perp$.
- (b) Show that $V^\perp$ is a subspace in $\mathbb{R}^n$.
- (c) Show that $V \cap V^\perp = \{\vec{0}\}$.
- (d) Show that $(V^\perp)^\perp = V$.

Question 3.12: Let M be an $n \times n$ -matrix with row vectors, $r_1, \dots, r_n$. Let,

$$\text{Row}(M) = \text{span}\{r_1, \dots, r_n\}$$

Show that,

$$\ker(M) = \text{Row}(M)^\perp$$

Question 3.13: Compute the row space of the following matrices,

$$\text{i) } M = \begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & -1 \\ 0 & 6 & -6 \end{bmatrix}, \quad \text{ii) } N = \begin{bmatrix} -2 & 0 & 8 \\ 6 & 27 & 3 \\ 4 & 16 & 0 \end{bmatrix}$$

Then compute $\ker(M)$ and show that it is orthogonal to $\text{Row}(M)$.

4 Worksheet 4: Matrices in higher dimensions

Question 4.1: Let $\vec{v} = [x_1, y_1, z_1]$, $\vec{w} = [x_2, y_2, z_2]$ be vectors in $\mathbb{R}^3$.

(a) Show that the vector,

$$\vec{v} \times \vec{w} = \begin{bmatrix} y_1 z_2 - y_2 z_1 \\ x_2 z_1 - x_1 z_2 \\ x_1 y_2 - x_2 y_1 \end{bmatrix}$$

has,

$$\vec{v} \cdot (\vec{v} \times \vec{w}) = 0$$

$$\vec{w} \cdot (\vec{v} \times \vec{w}) = 0$$

So it is orthogonal (at a right angle) to both $\vec{v}$ and $\vec{w}$. Draw a picture of this when $V = \text{span}\{\vec{v}, \vec{w}\}$ is a plane (i.e. when the vectors v, w don't lie on a line).

Use the formula above to see that if $v = \lambda w$ (so they lie both lie on a line) then,

$$\vec{v} \times \vec{w} = \vec{0}$$

- (b) Assume, $V = \text{span}\{\vec{v}, \vec{w}\}$ is a plane (so the vectors v, w are linearly independent). Use the picture from part (a) to convince yourself that if $\vec{r}$ is a vector in $\mathbb{R}^3$ which has,

$$\vec{r} \cdot (\vec{v} \times \vec{w}) = 0$$

then, $\vec{r} \in V = \text{span}\{v, w\}$ So $\vec{r}$ is in the plane spanned by $\vec{v}, \vec{w}$. This shows that,

$$\text{span}\{\vec{v} \times \vec{w}\} = V^\perp$$

In other words, $\text{span}\{\vec{v} \times \vec{w}\}$ is the line orthogonal to the plane $V = \text{span}\{\vec{v}, \vec{w}\}$ (and passes through the origin).

- (c) Let $\vec{i} = [1, 0, 0], \vec{j} = [0, 1, 0], \vec{k} = [0, 0, 1]$ (i, j, k is the convention in 3-dimensions instead of the the usual $b_1, \dots, b_n$). Show that,

$$\vec{i} \times \vec{j} = \vec{k}$$

$$\vec{j} \times \vec{k} = \vec{i}$$

$$\vec{k} \times \vec{i} = \vec{j}$$

- (d) Let,

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & i & j \end{bmatrix}$$

be a 3×3 matrix with column vectors $\vec{v}_1 = [a, d, g], \vec{v}_2 = [b, e, i], \vec{v}_3 = [c, f, j]$. Define its **determinant** as follows,

$$\det(M) = \vec{v}_1 \cdot (\vec{v}_2 \times \vec{v}_3)$$

Write this out and check that,

$$\det(M) = a \cdot \det\begin{pmatrix} e & f \\ i & j \end{pmatrix} - d \cdot \det\begin{pmatrix} b & c \\ i & j \end{pmatrix} + g \cdot \det\begin{pmatrix} b & c \\ e & f \end{pmatrix}$$

(e) Draw a picture and use part (b) to convince yourself that,

$$\det(M) \neq 0$$

only when the three column vectors v_1, v_2, v_3 are linearly independent (i.e. they point in genuinely different directions, not all of them lie on a plane or line).

(f) In the case where $\det(M) \neq 0$ (so the columns are linearly independent) draw a picture of the *parallelepiped* P bounded by the column vectors v_1, v_2, v_3 .

It turns out that,

$$\|\vec{v} \times \vec{w}\| = \|\vec{v}\| \|\vec{w}\| \sin(\theta)$$

which is the area of the parallelogram bounded by the vectors $\vec{v}, \vec{w}$ (just like in the plane!).

Use this to show that for a 3×3 -matrix,

$$|\det(M)| = \text{volume}(P) = \text{Area of base} \times \text{height} = \vec{v}_1 \cdot (\vec{v}_2 \times \vec{v}_3)$$

(g) Compute the determinant of the following matrices and determine whether the columns are linearly dependent or independent,

$$\text{i) } M = \begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & -1 \\ 0 & 6 & -6 \end{bmatrix}, \quad \text{ii) } N = \begin{bmatrix} -2 & 0 & 8 \\ 6 & 27 & 3 \\ 4 & 16 & 0 \end{bmatrix}$$

From n -dimensions to m -dimensions:

Question 4.2: Determine the image and kernel of the following matrices,

$$\text{i) } M = \begin{bmatrix} 1 & 2 & 10 & 0 \\ 4 & 3 & 2 & 1 \end{bmatrix}, \quad \text{ii) } N = \begin{bmatrix} 2 & 0 & 8 & 16 \\ 6 & 3 & 3 & 3 \\ 2 & 16 & 0 & 10 \end{bmatrix}$$

In each case show that, $\dim(\text{Im}(M)) + \dim(\ker(M)) = 4$

Define the **cokernel** of an $m \times n$ matrix as follows,

$$\text{coker}(M) = \{\vec{v} \in \mathbb{R}^n \mid \vec{v} \cdot \text{Im}(M) = 0\}$$

it is all the vectors orthogonal to $\text{Im}(M)$. If we have a basis $b_1, \dots, b_k$ for $\text{Im}(M)$ we can compute $\text{coker}(M)$ as the kernel of the matrix whose rows are the basis vectors $b_1, \dots, b_k$,

$$\text{coker}(M) = \ker \left(\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \end{bmatrix} \right)$$

In each of the cases from the previous question compute $\text{coker}(M)$ and show that,

$$\dim(\text{Im}(M)) + \dim(\text{coker}(M)) = \dim(\text{Range}) = 2 \text{ or } 3$$

Question: 4.3: Let M be an $m \times n$ matrix, so it is a map $M : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The **Rank-Nullity theorem** says,

$$\dim(\text{Im}(M)) + \dim(\ker(M)) = n$$

$$\dim(\text{Im}(M)) + \dim(\text{coker}(M)) = m$$

The dimension of $\text{Im}(M)$ is called the **rank** of M and the dimension of $\ker(M)$ is called the **nullity** of M .

(a) Show that,

$$n - m = \dim(\ker(M)) - \dim(\text{coker}(M))$$

(b) Show that if $n > m$ then,

$$\dim(\ker(M)) \geq 0$$

and if $n < m$ then,

$$\dim(\operatorname{coker}(M)) > 0$$

(c) Show that if, $\dim(\ker(M)) = 0$ then $m \geq n$ and if $\dim(\operatorname{Im}(M)) = n$ then $m \leq n$.

(d) Show that if $\dim(\ker(M)) = 0$ **and** $\dim(\operatorname{Im}(M)) = n$ then $n = m$.

Question 4.4: Compute the reduced row echelon form of the following matrices using *row operations*,

$$\text{i) } A = \begin{bmatrix} 1 & 0 & -6 & -1 \end{bmatrix}, \quad \text{ii) } B = \begin{bmatrix} -2 & -1 \\ -4 & 2 \end{bmatrix}, \quad \text{iii) } C = \begin{bmatrix} 1 & 4 & -1 & 3 \\ 2 & 9 & -1 & 7 \\ 2 & 8 & -2 & 6 \end{bmatrix}$$

In each case compute the rank and nullity of the matrix and note that,

$$\mathbf{rank} + \mathbf{nullity} = \mathbf{\#columns}$$

Inverse Matrices:

Question 4.4: Let M be a 3×3 -matrix.

(a) Write down matrices N such that the multiplication $N \cdot M$ performs the following row operations,

(i) Row 3 $\mapsto 2 \cdot \text{Row 1} + \text{Row 3}$

(ii) Row 2 $\mapsto 4 \cdot \text{Row 2}$

(iii) Row 2 $\mapsto \text{Row 2} - 3 \cdot \text{Row 1}$

(iv) Swap Row 1 and Row 3.

Compute the determinant of N in each case.

- (b) For each of the following elementary matrices, write out the row/column operations performed by left and right multiplication by N ,

$$\text{i) } M = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{ii) } O = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \quad \text{iii) } N = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 1 & 1 \\ 4 & 0 & 2 \end{bmatrix}$$

- (c) Show that if M is *upper triangular* or *lower triangular* (looks like the following matrices,

$$\text{upper triangular} = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}, \text{ lower triangular} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

then $\det(M)$ is the product of the entries along the diagonal. Check this in 2 and 3 dimensions!

- (d) Show that if E is the matrix representing the row operations,

(a) $R_1 \mapsto R_1 + \lambda R_2$, then $\det(E) = 1$

(b) $R_1 \mapsto \lambda R_1$, then $\det(E) = \lambda$

- (e) Use the fact that $\det(EM) = \det(E)\det(M)$ to show that if we can use row operations to get to reduce M to the identity matrix,

$$M \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

then $\det(M) \neq 0$.

- (f) Compute the determinant of each of the following matrices. Then try to compute the inverse matrix, and check that when $\det(M) = 0$ the process fails. If you are able to find an inverse, check that $M \cdot M^{-1} = \text{Id}$.

$$\text{i) } M = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 4 & -1 \\ 0 & 6 & -6 \end{bmatrix}, \quad \text{ii) } N = \begin{bmatrix} -2 & 0 & 8 \\ 6 & 27 & 3 \\ 4 & 16 & 10 \end{bmatrix}, \quad \text{iii) } O = \begin{bmatrix} -2 & 0 & 8 \\ 6 & 27 & 3 \\ 4 & 16 & 0 \end{bmatrix}$$

Question 4.5: An $n \times n$ matrix M is said to be **invertible** if there is another matrix M^{-1} such that,

$$M \cdot M^{-1} = M^{-1} \cdot M = \text{Id} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Let M be an invertible matrix. Show that every equation of the form,

$$M \cdot \vec{x} = \vec{v}$$

has a *unique* solution $\vec{x}$ (in other words there is a unique $\vec{x}$ such that $M \cdot \vec{x} = \vec{v}$). Do this in two steps,

(a) First use the inverse M^{-1} to write down a vector $\vec{x}$ such that,

$$M \cdot \vec{x} = \vec{v}$$

(b) Show that if $\vec{y}$ has,

$$M \cdot \vec{x} = M \cdot \vec{y} = \vec{v}$$

then,

$$M \cdot (\vec{x} - \vec{y}) = \vec{0}$$

(c) Then use the inverse to show that,

$$\vec{x} - \vec{y} = \vec{0}$$

$$\vec{x} = \vec{y}$$

Conclude that there is exactly one vector $\vec{x}$ such that,

$$M \cdot \vec{x} = \vec{v}$$

for any $\vec{v} \in \mathbb{R}^n$.

(d) Show that if M is invertible then,

$$\det(M^{-1}) = \frac{1}{\det(M)}$$

Question 4.6: Let M be an invertible matrix with columns $\vec{v}_1, \vec{v}_2, \vec{v}_3$. Let $B = \{b_1 = [1, 0, 0], b_2 = [0, 1, 0], b_3 = [0, 0, 1]\}$ be. Show that,

$$M^{-1} \cdot v_1 = b_1$$

$$M^{-1} \cdot v_2 = b_2$$

$$M^{-1} \cdot v_3 = b_3$$

5 Worksheet 5: Change of Basis

Question 5.1: Usually in $\mathbb{R}^3$ (or $\mathbb{R}^n$) we write vectors in terms of the **standard basis**,

$$\vec{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The defining feature of a basis for this question is that any vector $[x, y, z]$ has a unique expression

of the form,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Let $B = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ be another basis for $\mathbb{R}^3$. So again every vector $\vec{v} = [x, y, z]$ has a unique expression,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a \cdot \vec{v}_1 + b \cdot \vec{v}_2 + c \cdot \vec{v}_3$$

- (a) Let B be the matrix whose columns form a basis $B = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. Show that there is a matrix, B^{-1} such that,

$$B^{-1} \cdot \vec{v}_1 = b_1$$

$$B^{-1} \cdot \vec{v}_2 = b_2$$

$$B^{-1} \cdot \vec{v}_3 = b_3$$

and conclude that,

$$B \cdot B^{-1} = B^{-1} \cdot B = \text{Id} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

.

Note: We saw in the last worksheet that if a matrix M is invertible, then for every vector $\vec{v}$ the equation,

$$M \cdot \vec{x} = \vec{v}$$

has a unique solution $\vec{x}$. This is the same as saying its columns form a basis.

- (b) Let $\vec{v} = [x, y, z]$ be a vector in $\mathbb{R}^3$ and let B be a matrix whose columns form a basis B as

above. We define the **coordinate vector** of $\vec{v}$ in terms of B as follows,

$$\vec{v}_B = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = B^{-1} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

In other words, $\vec{v}_B$ is the unique vector such that,

$$B \cdot \vec{v}_B = a \cdot \vec{v}_1 + b \cdot \vec{v}_2 + c \cdot \vec{v}_3 = \vec{v}$$

Compute the coordinate vectors of $\vec{v} = [1, 2, 3]$, $\vec{w} = [1, 5, 7]$ in terms of the following bases,

$$(a) \ B_1 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \right\}$$

$$(b) \ B_2 = \left\{ \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

In each case check your answer by computing $M \cdot \vec{v}_B = \vec{v}$.

Question 5.2: We have seen in the past how to write down matrices which perform simple operations on the standard basis vectors.

For instance check that the matrix,

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

cycles the three standard basis vectors,

$$b_1 \mapsto b_2 \mapsto b_3 \mapsto b_1$$

and check that $M^3 = \text{Id}$.

(a) What matrix has the following effect on the standard basis,

$$b_1 \mapsto b_1 - b_3$$

$$b_2 \mapsto b_2 + 3b_3$$

$$b_3 \mapsto b_3$$

(b) What matrix does this?

$$b_1 \mapsto 3b_2$$

$$b_2 \mapsto 2b_1$$

$$b_3 \mapsto b_1 + b_2 + b_3$$

(c) Let B be a 3×3 matrix whose columns form a basis $B = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and let,

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Let,

$$M_B := B^{-1} \cdot M \cdot B$$

Show that,

$$M_B \cdot [\vec{v}_i]_B = [M \cdot \vec{v}_i]_B$$

Compute the matrices,

$$\text{i) } M = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \text{ii) } N = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

in terms of the basis B_1 from question 5.1. Show that M_B cycles the vectors,

$$[b_1]_B \rightarrow [b_2]_B \rightarrow [b_3]_B \rightarrow [b_1]_B$$

and N_B scales the vectors,

$$[b_1]_B, [b_2]_B, [b_3]_B$$

by the constants 1, 2, 3.

Question 5.3: Show that in 2- dimensions the matrix,

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

has the effect of flipping the (x, y) -plane across the line $y = x$ (which is the span of the vector $[1, 1]$).

(a) Let $v_1 = [1, 1]$, $v_2 = [1, -1]$. Check that,

$$M \cdot v_1 = v_1$$

$$M \cdot v_2 = -v_2$$

(b) Let B be the matrix whose columns are the vectors v_1, v_2 . Show that,

$$M_B = B^{-1}MB = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Then show that,

$$M_B \cdot v_1 = v_2$$

$$M_B \cdot v_2 = v_1$$

and that,

$$M_B \cdot [\vec{v}_i]_B = [M \cdot \vec{v}_i]_B$$

(c) Let M be a 2×2 matrix such that $M^2 = \text{Id}$. Show that M is of the form,

$$M = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

where $a^2 + bc = 1$.

Question 5.3: A standard ellipse in the (x, y) -plane is cut out by an equation of the form,

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

(a) Open the Desmos online graphing calculator and graph the above equation, experiment with the values a, b to see how they affect the picture.

(b) Let $M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $M_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Show that, if $\vec{v} = [x, y]$ is a solution to the equation,

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

then so are $M_1 \cdot \vec{v}$ and $M_2 \cdot \vec{v}$.

(c) Let $\vec{m} = [1, m]$ so that,

$$L = \text{span}\{[1, m]\}$$

is the line passing through the origin with slope m .

(a) Show that the vector $\vec{m}^\perp = [m, -1]$ is orthogonal to $\vec{m}$ and that the vectors $B = \{\vec{m}, \vec{m}^\perp\}$ form a basis for $\mathbb{R}^2$. (Show that the matrix with B as its columns has $\det \neq 0$)

(b) Write down an equation for an ellipse which is oriented along the line L as follows,

i. Compute the coordinate vector for $\vec{v} = [x, y]$ in terms of the basis B .

$$[\vec{v}]_B = B^{-1} \cdot \vec{v} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

ii. Substitute the expressions x', y' into the equation,

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

to obtain a new equation. Choose values like $a = 2, b = 3, m = 4$ and graph the resulting equation on desmos to see that the curve is again an ellipse but rotated.

Question 5.4: A parabola is cut out by the equation,

$$x^2 - y = 0$$

(a) Use the following matrices to change coordinates,

$$\text{i) } B_1 = \begin{bmatrix} 1 & 2 \\ 0 & 5 \end{bmatrix}, \quad \text{ii) } B_2 = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}, \quad \text{iii) } B_3 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

In each case, write out the resulting equation and draw a picture of the curve it cuts out in the (x, y) -plane. In one of the cases above you should get a circle again, what is special about the basis vectors in that matrix?

(b) Write down a matrix which preserves the solutions of,

$$x^2 - y = 0$$

(c) Use the same matrices to change coordinates for the equation,

$$x^2 + y^2 - 1 = 0$$

again write out the resulting equations and draw a picture of the curve in the (x, y) -plane

(d) Let M be an orthogonal matrix. So $M \cdot M^T = M^T \cdot M = \text{Id}$. Show that if $\vec{v} = [x, y]$ is a solution to the equation,

$$x^2 + y^2 - 1 = 0$$

then so is $M \cdot \vec{v}$. Hint: $(M\vec{v})^T = \vec{v}^T M^T$ and,

$$||\vec{v}||^2 = x^2 + y^2 = \vec{v}^T \vec{v} = \vec{v} \cdot \vec{v}$$

6 Worksheet 6: Eigenvectors and Diagonalization

Question 6.1: The eigenvalues of an $n \times n$ -matrix M are the values of λ for which $\det(M - \lambda \cdot \text{Id}) = 0$. If we replace λ by a variable X we get,

$$f_M(X) = \det(M - X \cdot \text{Id})$$

called the **characteristic polynomial** of M . The roots of this polynomial are the eigenvalues of M .

Let M be the matrix,

$$M = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Find all the eigenvalues of M and the corresponding eigenvectors. Show that the eigenvectors of M form a basis for $\mathbb{R}^3$.

Question 6.2:

(a) Let M be a 2×2 -matrix,

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Show that,

$$f_M(X) = \det(M - X \cdot \text{Id})$$

is given by the polynomial,

$$f_M(X) = X^2 - (a + d)X + (ad - bc)$$

Notice the determinant $\det(M) = ad - bc$ shows up as the constant term! The X coefficient,

$$\text{Tr}(M) = a + d$$

is called the **trace** of M , which is the sum of the entries along the main diagonal.

- (b) Let λ_1, λ_2 be the roots of the polynomial $f_M(X)$, in other words they are the eigenvalues of M . Use the quadratic formula to show that,

$$\text{Tr}(M) = a + d = \lambda_1 + \lambda_2$$

and,

$$\det(M) = ad - bc = \lambda_1 \lambda_2$$

- (c) Use $f_M(X)$ to show that 0 is an eigenvalue of M if and only if $\det(M) = 0$. In other words,

$$\ker(M) \neq \{\vec{0}\} \iff \det(M) = 0$$

- (d) **Cayley-Hamilton Theorem:** You can check that,

$$f_M(M) = M^2 - (a + d)M + (ad - bc) \cdot \text{Id} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

in other words plugging M into its characteristic polynomial gives the zero matrix.

- (e) Let N be an invertible matrix. Show that if v is an eigenvector of M with eigenvalue λ then $N^{-1}v$ is an eigenvector for,

$$N^{-1}MN$$

with the same eigenvalue. Similarly if w is an eigenvector for $N^{-1}MN$ then Nw is an eigenvector for M .

(Up to scaling) Two polynomials are the same $\iff$ they have the same roots (with the same multiplicity). Use this to argue that,

$$f_M(X) = f_{N^{-1}MN}(X)$$

Question 6.3:

(a) Let M be the matrix,

$$M = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

use the polynomial $f_M(X)$ to compute the eigenvalues of M . Then compute, $\ker(M - \lambda \text{Id})$ to find the eigenvectors of M . There should be two eigenvalues, $\pm\lambda$ and two eigenvectors $\vec{v}_+, \vec{v}_-$. Let N be the matrix,

$$N = [\vec{v}_+, \vec{v}_-]$$

with the $\vec{v}_\pm$ as columns. Show that,

$$MN = N \cdot \begin{bmatrix} \lambda & 0 \\ 0 & -\lambda \end{bmatrix}$$

and hence that,

$$N^{-1}MN = \begin{bmatrix} \lambda & 0 \\ 0 & -\lambda \end{bmatrix}$$

(d) Show that the eigenvalues of,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

are $i = \sqrt{-1}$ and $-i = -\sqrt{-1}$. (imaginary numbers!)

So the matrix,

$$M - i \cdot \text{Id}$$

has entries which are complex numbers. Compute,

$$\ker(M - (\pm i) \cdot \text{Id})$$

notice that this space lives in $\mathbb{C}^2$ and not in $\mathbb{R}^2$.

Question 6.4: Let M be a 2×2 matrix such that $M^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

- (a) Show that $\vec{0}$ is the only eigenvalue of M . Hint: Show that if λ is an eigenvalue of M then λ^2 is an eigenvalue of M^2 .

This shows that $f_M(X) = X^2$ without having to compute any determinants!

- (b) Show that if $M \neq [0]$ then $\ker(M)$ is a line in $\mathbb{R}^2$. Then show that $\text{Im}(M) = \ker(M)$.
- (c) Choose a nonzero vector $\vec{v}$ in $\ker(M) = \text{Im}(M)$ and a vector $\vec{v} \neq \vec{w}$ such that,

$$\vec{v} = M\vec{w}$$

Show that $\vec{v}, \vec{w}$ don't lie on the same line and therefore $B = \{\vec{v}, \vec{w}\}$ is a basis for $\mathbb{R}^2$.

- (d) Let N be the matrix with columns $N = [\vec{v}, \vec{w}]$. Show that,

$$N^{-1}MN = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Question 6.5: Let M be the matrix,

$$M = \begin{bmatrix} a & 0 & b \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$$

- (a) Compute the polynomial,

$$f_M(X) = \det(M - X \cdot \text{Id})$$

and show that a is the only eigenvalue.

(b) Show that $\ker(M - a \cdot \text{Id})$ is a plane in $\mathbb{R}^3$ and compute a basis for this plane.

Question 6.6: Fibonacci Sequence: Let M be the matrix,

$$M = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

(a) The Fibonacci Sequence is the sequence which starts $a_0 = 0, a_1 = 1$ then,

$$a_{n+1} = a_n + a_{n-1}$$

Use induction to show that,

$$M^n \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a_{n+1} \\ a_n \end{bmatrix}$$

(b) Compute the polynomial,

$$f_M(X) = \det(M - X \cdot \text{Id})$$

and show the eigenvalues of M are given by,

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}, \lambda_2 = \frac{1 - \sqrt{5}}{2}$$

λ_1 is called the **golden ratio**

(c) Compute the eigenvectors $\vec{v}_1, \vec{v}_2$ associated to the eigenvalues λ_1, λ_2 . Let N be the matrix with columns $\vec{v}_1, \vec{v}_2$. Show that,

$$N^{-1}MN = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

then show that,

$$N^{-1}M^nN = \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix}$$

(d) Use induction to show that,

$$M^n \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\lambda_1^n - \lambda_2^n}{\lambda_1 - \lambda_2} \\ \frac{\lambda_1^{n-1} - \lambda_2^{n-1}}{\lambda_1 - \lambda_2} \end{bmatrix}$$

Then show,

$$\frac{a_n}{a_{n-1}} = \frac{\lambda_1^n - \lambda_2^n}{\lambda_1^{n-1} - \lambda_2^{n-1}}$$

(e) Show that in the limit,

$$\lim_{n \rightarrow \infty} \lambda_2^n = 0$$

so,

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = \lambda_1 = \frac{1 + \sqrt{5}}{2}$$

Question 6.7: Let M be an $n \times n$ matrix and let $\lambda_1 \neq \lambda_2$ be distinct eigenvalues of M , with corresponding eigenvectors $\vec{v}_1, \vec{v}_2$.

(a) Suppose that the eigenvectors $\vec{v}_1, \vec{v}_2$ are linearly dependent. So we can write,

$$x_1 \vec{v}_1 + x_2 \vec{v}_2 = \vec{0}$$

where $x_1, x_2 \neq 0$. Apply the matrix, $M - \lambda_1 \cdot \text{Id}$ to the above expression to see that,

$$(M - \lambda_1 \text{Id})(x_1 \vec{v}_1 + x_2 \vec{v}_2) = x_2(\lambda_2 - \lambda_1)\vec{v}_2 = \vec{0}$$

Argue that this can only be true if $\lambda_1 = \lambda_2$ which contradicts our assumption. Conclude that the $\vec{v}_1, \vec{v}_2$ are linearly independent. (i.e. they don't lie on the same line)

Question 6.8: We say that an $n \times n$ matrix A is **symmetric** if,

$$A^T = A$$

(a) Let $\lambda_1 \neq \lambda_2$ be distinct eigenvalues of a symmetric matrix A , and let $\vec{v}_1, \vec{v}_2$ be eigenvectors

corresponding to these eigenvalues. Use the identity,

$$(A\vec{x})^T = \vec{x}^T A^T$$

to show that,

$$\vec{v}_1 \cdot \vec{v}_2 = \vec{v}_1^T \vec{v}_2 = 0$$

This shows that the eigenspaces of a symmetric matrix are all orthogonal to each-other

(b) Every complex number $z \in \mathbb{C}$ can be written,

$$z = a + bi$$

where $a, b \in \mathbb{R}$. We write,

$$\bar{z} = a - bi$$

for its **conjugate**. Show that,

$$z = \bar{z} \iff z = a + 0i \in \mathbb{R}$$

(c) Show that if $\vec{v} \in \mathbb{C}^n$ then $\vec{v}$ can be written,

$$\vec{v} = \vec{a} + i\vec{b}$$

with $\vec{a}, \vec{b} \in \mathbb{R}^n$. We write $\bar{\vec{v}} = \vec{a} - i\vec{b}$ for its conjugate.

(d) Show that,

$$\bar{\vec{v}} \cdot \vec{v} = ||\vec{a}||^2 + ||\vec{b}||^2$$

and,

$$\bar{\vec{v}} \cdot \vec{v} = 0 \iff \vec{v} = \vec{0}$$

(e) **Symmetric matrices have real eigenvalues:** Let A be an $n \times n$ symmetric matrix with real

entries and let $\lambda \in \mathbb{C}$ be an eigenvalue of A . So,

$$A\vec{v} = \lambda\vec{v}$$

for some nonzero vector $\vec{v} \in \mathbb{C}^n$. Show that,

$$A\bar{v} = \bar{\lambda}\bar{v}$$

Then use the fact that $A = A^T$ to show,

$$\bar{v}^T A v = \bar{v}^T (\lambda v) = \lambda(\bar{v} \cdot v)$$

$$\bar{v}^T A v = (A\bar{v})^T \vec{v} = \bar{\lambda}(\bar{v} \cdot \vec{v})$$

Since $\vec{v} \neq \vec{0}$, we can divide by $(\bar{v} \cdot \vec{v})$ to get,

$$\bar{\lambda} = \lambda$$

which we saw means $\lambda \in \mathbb{R}$

Question 6.9: Recall that a matrix M is **orthogonal** if,

$$MM^T = MM^T = \text{Id}$$

We saw in 2-dimensions these correspond to rotations (or reflections) in the plane. This turns out to be true in n -dimensions as well. In this question we will show that every rotation (or reflection) in n -dimensions where n is odd, has a fixed axis.

- (a) It turns out that for any $n \times n$ matrix, $\det(M) = \det(M^T)$, and for any two $n \times n$ matrices, $\det(MN) = \det(M) \cdot \det(N)$. Use this to show that if M is orthogonal then,

$$\det(M) = \pm 1$$

- (b) The dot product of a vector with itself measures its length,

$$\vec{v} \cdot \vec{v} = \vec{v}^T \vec{v} = \|\vec{v}\|^2$$

Use the formula $(M\vec{v})^T = \vec{v}^T M^T$ to show that if M is orthogonal then,

$$\|M\vec{v}\| = \|\vec{v}\|$$

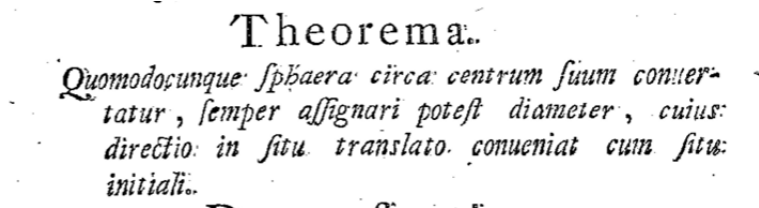
- (c) Use part (b) to argue that if $\lambda \in \mathbb{C}$ is an eigenvalue of an orthogonal matrix M then,

$$|\lambda| = 1$$

so all the eigenvalues of M lie on the unit circle in $\mathbb{C}$.

- (d) Show that if $\lambda \in \mathbb{C}$ is an eigenvalue of M then so is $\bar{\lambda}$. Use this to argue that if n is odd and M is an orthogonal matrix then there is at least one eigenvalue such that, $\lambda = \bar{\lambda}$ (which implies this eigenvalue λ is a real number.)
- (e) Use parts (c), (d) to argue that if n is odd then $\lambda = \pm 1$ is an eigenvalue. We get the following result of Euler in the case $n = 3$,

Euler's Rotation Theorem (1775):



A screenshot of the original theorem from the Euler archives,

<https://scholarlycommons.pacific.edu/euler-works/478/>

In English: When a sphere is moved around its centre it is always possible to find a diameter whose direction in the displaced position is the same as in the initial position.

- (f) Go back to Worksheet 2 and write down a counterexample to the above argument in 2-dimensions. You should find a 2×2 orthogonal matrix M that does not have ± 1 as an

eigenvalue. Can you use M to give counter-examples in dimensions $n = 4, 6, 8, 12, 14, \dots$ etc?

7 Worksheet 7: Vector Spaces and Linear Transformations

Question 7.1: Every complex number $z \in \mathbb{C}$ can be written,

$$z = a + bi$$

where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$.

- (a) Show that $\mathbb{C}$ is a vector space over $\mathbb{R}$ and $\dim_{\mathbb{R}}(\mathbb{C}) = 2$, what is a nice basis for this vector space?
- (b) Show that for every $z \neq 0 \in \mathbb{C}$ the multiplication by z map,

$$\text{mult}_z : \mathbb{C} \rightarrow \mathbb{C}$$

is a linear isomorphism. What is the inverse of this linear map?

- (c) Write down the matrix representation of mult_z with respect to your basis from part (a). Then compute its characteristic polynomial and show that $z = a + bi$ and $\bar{z} = a - bi$ are the eigenvalues.
- (d) What is the matrix representation for mult_i ? Describe the effect this linear transformation has on the plane.

Question 7.2: Let V be the vector space of all functions of the form,

$$V = \{f(X) = a \sin(X) + b \cos(X) \mid a, b \in \mathbb{R}\}$$

Describe a basis for V and show that differentiation with respect to X gives a linear map,

$$\frac{\partial}{\partial X} : V \rightarrow V$$

Write down the matrix representation of $\partial/\partial X$ with respect to your basis.

Question 7.3: Let,

$$\mathbb{R}[X]_k = \{f(X) = a_0 + a_1X + \cdots + a_kX^k \mid a_0, \dots, a_k \in \mathbb{R}\}$$

be the set of all polynomials with real coefficients and degree $\leq k$.

(a) Show that $\mathbb{R}[X]_k$ is a vector space and describe a basis B for this space.

(b) We write $\frac{\partial}{\partial X}$ for the derivative with respect to X . Show that the derivative gives a linear map,

$$\frac{\partial}{\partial X} : \mathbb{R}[X]_k \rightarrow \mathbb{R}[X]_k$$

Describe the kernel and image of $\frac{\partial}{\partial X}$.

(c) Write down the matrix representation of the derivative with respect to the basis B from part (a).

(d) Write ev_0 for the *evaluation map*,

$$\begin{aligned} ev_0 : \mathbb{R}[X]_k &\rightarrow \mathbb{R} \\ f(X) &\mapsto f(0) \end{aligned}$$

Show that this is a linear map, and that it is surjective but not injective. What is the kernel?

Question 7.4: Let X be a finite set and let,

$$\mathbb{R}(X) := \{\text{functions } f : X \rightarrow \mathbb{R}\}$$

(a) Show that $\mathbb{R}(X)$ is a vector space under the operations,

$$[f + g](x) = f(x) + g(x)$$

$$[\lambda f](x) = \lambda f(x)$$

What is the zero vector in $\mathbb{R}(X)$?

(b) Describe a basis $B = \{f_x \mid x \in X\}$ for $\mathbb{R}(X)$ indexed by the elements of X . Prove that it is a basis by showing,

1. $\text{span}(B) = \mathbb{R}(X)$, so every function f can be written as a combination,

$$f = \lambda_1 f_{x_1} + \cdots + \lambda_n f_{x_n}$$

2. If you remove any basis function f_x , then $\text{span}(B - \{f_x\}) \neq \mathbb{R}(X)$. So all the f_x are necessary to span $\mathbb{R}(X)$

(c) Let Y be another finite set and let π be a function,

$$\pi : Y \rightarrow X$$

Use π to define a function,

$$\pi^* : \mathbb{R}(X) \rightarrow \mathbb{R}(Y)$$

and show that π^* is a linear map.

(d) Let $X = \{1, 2, 3\}$ and let $\pi : X \rightarrow X$ be the function,

$$\pi := \begin{cases} 1 \mapsto 1 \\ 2 \mapsto 1 \\ 3 \mapsto 2 \end{cases}$$

Where does π^* send the basis $B = \{f_x \mid x \in X\}$? Use this to write down the matrix representation of $[\pi^*]_B$. Compute the kernel and image of $[\pi^*]_B$.

- (e) When is the linear map π^* is an isomorphism (surjective and injective)? Try to guess the answer and prove it!
-

Question 7.5: Let $V = \text{Mat}_2(\mathbb{R})$ be the space of real 2×2 matrices.

- (a) Show that V is a vector space. Describe a basis for V and compute $\dim(V)$.
- (b) Let $W = \{M \in V \mid \text{Tr}(M) = 0\}$ be the collection of trace zero matrices in V . Show that W is a subspace of V and that, $E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, F = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ forms a basis for W . Conclude that $\dim(W) = 3$.
- (c) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ let $W' = \text{span}(A)$. Show that every matrix $M \in V$ can be written as a sum,

$$M = N + R$$

where $N \in W$ and $R \in W'$.

Question 7.6: Let V, W be vector spaces and let $\varphi : V \rightarrow W$ be a linear map.

- (a) Show that φ is injective if and only if $\ker(\varphi) = \{\vec{0}\}$.
- (b) Suppose we have a collection of vectors $v_1, \dots, v_n$ that span V . Show that φ is surjective if and only if the $\varphi(v_1), \dots, \varphi(v_n)$ span W .
- (c) Suppose the $v_1, \dots, v_n$ form a basis for V . Show that φ is an isomorphism if and only if $\varphi(v_1), \dots, \varphi(v_n)$ is a basis for W .
-

Question 7.7: Let V be a vector space with basis $B = \{b_1, \dots, b_n\}$ (so $\dim(V) = n$ is finite).

(a) Use B to define a linear map,

$$\varphi : \mathbb{R}^n \rightarrow V$$

show that φ is an isomorphism.

(c) **The Dual Space:** Let,

$$V^* := \{f : V \rightarrow \mathbb{R} \mid f \text{ linear}\}$$

we call V^* the dual space of V . Use B to describe a basis for V^* and conclude that $\dim(V) = \dim(V^*)$

(d) Let $V = \mathbb{R}^n$ and let,

$$\langle v, w \rangle := \vec{v} \cdot \vec{w}$$

denote the usual dot product. For each vector $\vec{v}$ we get an element of dual space,

$$\begin{aligned} \langle \vec{v}, - \rangle : V &\rightarrow \mathbb{R} \\ \vec{w} &\mapsto \langle \vec{v}, \vec{w} \rangle \end{aligned}$$

i.e. a linear map $V \rightarrow \mathbb{R}$. Show that,

$$\begin{aligned} V &\rightarrow V^* \\ \vec{v} &\mapsto \langle \vec{v}, - \rangle \end{aligned}$$

is a linear isomorphism. Hint: Show that $\ker = \{\vec{0}\}$, so the map is injective, then use part (c) to conclude that it is surjective as well.

(e) **Double dual:** Let $(V^*)^*$ be the dual space to the dual of V and consider the map,

$$\begin{aligned} V &\rightarrow (V^*)^* \\ \vec{v} &\mapsto \phi_v \end{aligned}$$

where ϕ_v is the linear map, $\phi_v : V^* \rightarrow \mathbb{R}$ such that,

$$\phi_v(f) = f(v)$$

Show that the map $v \mapsto \phi_v$ gives a linear isomorphism $V \cong (V^*)^*$. Note that here we didn't have to choose a basis!

Question 7.8: Let V be a vector spaces and let $W \subset V$ be a subspace (i.e. a subset which is also a vector space). Define an equivalence relation $\sim$ on V by,

$$\vec{v}_1 \sim \vec{v}_2 \iff \vec{v}_1 - \vec{v}_2 \in W$$

- (a) Show that $\sim$ is an equivalence relation, so it is reflexive, symmetric and transitive.
- (b) Show that the equivalence class of $\vec{v} \in V$ is given by,

$$[\vec{v}] = \vec{v} + W = \{\vec{v} + \vec{w} \mid \vec{w} \in W\}$$

for some $\vec{v}$, and in particular $[\vec{0}] = W$. Think of these equivalence classes as "shifted" versions of the subspace W .

- (c) Let $V = \mathbb{R}^2$ and $W = \text{span} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Draw a picture of the equivalence classes for $\sim$ in $\mathbb{R}^2$.
- (d) Write V/W for the set of equivalence classes of $\sim$. Show that V/W is a vector space with the operations,

$$[\vec{v}_1] + [\vec{v}_2] = [\vec{v}_1 + \vec{v}_2]$$

$$\lambda[\vec{v}] = [\lambda\vec{v}]$$

- (e) **The First Isomorphism Theorem:** Let $\varphi : V \rightarrow W$ be a linear map. For free, φ gives us a map,

$$\tilde{\varphi} : V / \ker(\varphi) \rightarrow \text{Im}(\varphi)$$

which sends the equivalence class,

$$[\vec{v}] = \vec{v} + \ker(\varphi) \mapsto \varphi(\vec{v})$$

- (a) Show that this map is linear and well defined. Well defined means that whenever $[\vec{v}_1] = [\vec{v}_2] \in V/\ker(\varphi)$ we have $\varphi(\vec{v}_1) = \varphi(\vec{v}_2)$.
- (b) Show that $\tilde{\varphi}$ is surjective (if $\varphi(\vec{v}) \in \text{Im}(\varphi)$ what equivalence class maps to it?) and
- (c) Show that $\tilde{\varphi}$ is injective (show that if $\varphi(\vec{v}_1) = \varphi(\vec{v}_2)$ then $[\vec{v}_1] = [\vec{v}_2] \in V/\ker(\varphi)$). Conclude that $\tilde{\varphi}$ is an isomorphism.

It turns out that $\dim(V/\ker(\varphi)) = \dim(V) - \dim(\ker(\varphi))$, and you can use this to prove the Rank-Nullity theorem. We have shown above that

$$\dim(V/\ker(\varphi)) = \dim(V) - \dim(\ker(\varphi)) = \dim(\text{Im}(\varphi))$$

which shows,

$$\dim(V) = \dim(\ker(\varphi)) + \dim(\text{Im}(\varphi))$$

Question 7.9: Transpose: Let $\varphi : V \rightarrow W$ be a linear map between finite dimensional vector spaces (say $\dim(V) = n$ and $\dim(W) = m$). We define its transpose,

$$\varphi^T : W^* \rightarrow V^*$$

$$\varphi(f) = f \circ \varphi$$

where $f \in W^*$.

- (a) Check that $\varphi(f) \in V^*$, i.e. that it is a linear map $V \rightarrow \mathbb{R}$.
- (b) Write $\text{Hom}(V, W) := \{\varphi : V \rightarrow W \mid \varphi \text{ linear}\}$ for the set of all linear maps from V to W . Show that $\text{Hom}(V, W)$ is a vector space and that the transpose gives a linear map,

$$(-)^T : \text{Hom}(V, W) \rightarrow \text{Hom}(W^*, V^*)$$

(c) What is the dimension of $\text{Hom}(V, W)$? Use question 7.7 to conclude that,

$$\dim(\text{Hom}(V, W)) = \dim(\text{Hom}(W^*, V^*))$$

(d) Show that the transpose map is injective by showing it has $\ker((-)^T) = \vec{0}$ where $\vec{0}$ is the linear map which sends all of V to $\vec{0}$ in W .

(e) Use parts c, d to conclude that,

$$(-)^T : \text{Hom}(V, W) \rightarrow \text{Hom}(W^*, V^*)$$

is an isomorphism.

8 Projects

8.1 Lattice Based Cryptography:

Lattice-based cryptography is being pushed as a future alternative to modern day RSA encryption schemes, which use prime numbers. Currently, almost all computers in the world use RSA to encrypt peoples private info. Quantum computers will soon have the ability to crack these RSA encryption schemes, so people are looking for new alternatives which can withstand attacks by quantum computers. Here is a longer reference, <https://core.ac.uk/download/639166364.pdf>, although youtube videos and summaries are available online.

Let $B = \{\vec{b}_1, \dots, \vec{b}_n\}$ be a basis for $\mathbb{R}^n$. In other words, $\text{span}(B) = \mathbb{R}^n$ and every vector $\vec{v} \in \mathbb{R}^n$ has a *unique* expression,

$$\vec{v} = x_1\vec{b}_1 + x_2\vec{b}_2 + \dots + x_n\vec{b}_n$$

where the scalars $x_i \in \mathbb{R}$ are *real numbers*.

If we restrict ourselves and require that the x_i are *integers*, then we get what's called a **lattice** in $\mathbb{R}^n$.

$$L = \text{span}_{\mathbb{Z}}(B) = \{x_1 \vec{b}_1 + x_2 \vec{b}_2 + \cdots + x_n \vec{b}_n \mid x_i \in \mathbb{Z}\}$$

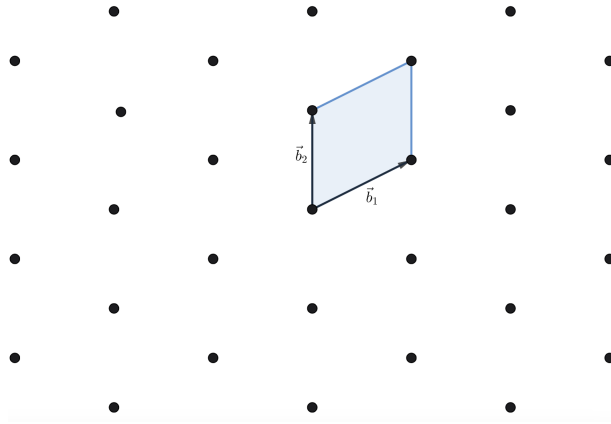


Figure 1: A picture of a lattice in $\mathbb{R}^2$, and a "good" basis

The **GGH encryption scheme** is a way to encrypt data using a lattice. The rough idea is as follows,

- (a) Alice wants to send Charlotte a message, but doesn't want Bob to know what it says. The message $\vec{m} = [m_1, \dots, m_n]$ is encoded as a vector in $\mathbb{Z}^n$.

Charlotte has a lattice L in mind, and a *good* basis B for the lattice. So the vectors in B are relatively short and as close to orthogonal as they can be. Write R for the matrix with the basis vectors B as its columns. This is her **private key**.

Alice and Bob have access to the lattice L via a **public key** in the form of a *bad* basis B' (really as a matrix $R' = RK$ where K is a matrix with $\det(K) = 1$). Imagine that the vectors in B' are close together and relatively long. In practice (i.e., in very high dimensions like $n = 15000$), this makes it difficult for a computer to determine where things lie on the lattice.

(b) Alice sends the message $\vec{m}$ to Charlottes lattice by taking the product,

$$R' \cdot \vec{m}$$

Then to throw others off she adds a bit of error to the message,

$$\vec{m}_{enc} = R' \cdot \vec{m} + \vec{e}$$

where $\vec{e}$ is a very small vector in a random direction. If Bob wants to figure out what $\vec{m}$ was, he has to determine the coordinates of the vector closest to $\vec{m}_{enc}$ in the lattice, with the public "bad" basis, this is extremely difficult to do. Charlotte can do this relatively easily with her private key. First she computes,

$$R^{-1} \cdot \vec{m}_{enc} = K \cdot \vec{m} + R^{-1} \cdot \vec{e}$$

then she finds the vector, $\vec{m}$ by rounding the coordinates $R^{-1} \cdot \vec{m}_{enc}$ to their nearest whole number entries.

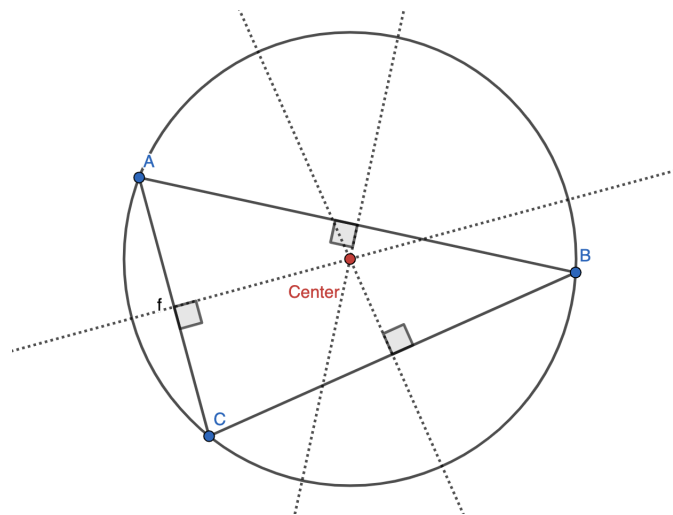
8.2 Curve Interpolation Problems:

I learned about these sorts of problems in a really great talk by the mathematician Isabel Vogt! see here <https://www.youtube.com/watch?v=shoqpbAL06o&t=1159s>.

The following is a theorem of Euclid (the Euclid we are referring to when we say Euclidean space - $\mathbb{R}^n$!),

Theorem (Euclid): Given three points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ in the plane (not lying on a line), there is a unique circle which passes through all three points.

For a given three points, Euclid actually gives a way to construct the circle using a ruler and compass! The picture is as follows,



You can read about this more online, but the amazing thing is that problems like this can be proved using linear algebra!

Here is another theorem proved by a Greek mathematician Pappus (said in modern terms),
Theorem (Pappus): Let $(x_1, y_1), \dots, (x_5, y_5)$ be 5 ("relatively spread out") points in the plane. Then there is a unique *conic* which passes through them. A *conic* is an equation of the form,

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

where a, b, c, d, e, f are real numbers. Try graphing some of these in desmos to get a sense of what

they look like! For instance,

$$x^2 + y^2 - 1 = 0$$

is the equation of the unit circle, but conics are more general.

Here is a proof that uses linear algebra (taken from Prof. Vogt's talk),

Proof. We won't show the conic is unique, only that it exists. To find the conic, we are looking for numbers a, b, c, d, e, f such that all 5 of the points (x_i, y_i) satisfy the defining equation,

$$ax_1^2 + bx_1y_1 + cy_1^2 + dx_1 + ey_1 + f = 0$$

$$ax_2^2 + bx_2y_2 + cy_2^2 + dx_2 + ey_2 + f = 0$$

$$ax_3^2 + bx_3y_3 + cy_3^2 + dx_3 + ey_3 + f = 0$$

$$ax_4^2 + bx_4y_4 + cy_4^2 + dx_4 + ey_4 + f = 0$$

$$ax_5^2 + bx_5y_5 + cy_5^2 + dx_5 + ey_5 + f = 0$$

but we can write this as a matrix equation of the form,

$$M = \begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Which is to say we need a vector $[a, b, c, d, e, f]$ in the kernel of the above matrix! Note that the matrix M has 6 columns and 5 rows, so it gives us a map,

$$M : \mathbb{R}^6 \rightarrow \mathbb{R}^5$$

since 6 is bigger than 5, the Rank-Nullity theorem implies there must be a vector $[a, b, c, d, e, f]$ in $\ker(M)$. This vector gives us the conic we are looking for.

In fact even more is true. If the points chosen $(x_1, y_1), \dots, (x_5, y_5)$ are "random enough" then there will be a *unique* conic which passes through them, which is to say that, $\dim(\ker(M)) = 1$ $\square$

Experiment with this! Pick a few points in the plane with easy whole number coordinates, then compute the kernel of the above matrix (you can use an online calculator if you like). You can then graph the conic you get and check that it passes through all the points!

This argument can be used to prove a theorem from much later by a Swiss mathematician Cramer (from Cramer's rule!),

Theorem (Cramer): Given $d(d+3)/2$ points in the plane, there is a degree d curve which passes through all of them.

Watch at least the first half of Prof. Vogt's talk to see her explain these and a few other similar problems!

8.3 Matrix Exponentials and Linear Differential Equations:

The **simple harmonic oscillator** is a system of two differential equations,

$$\frac{\partial x}{\partial t} = -y(t), \frac{\partial y}{\partial t} = x(t)$$

We would like to solve these equations. In other words, can we find functions $x(t), y(t)$ whose derivatives satisfy the above equations? If we think of the pair $\vec{v}(t) = [x(t), y(t)]$ as a vector which moves around over time t , then we can write the above equations in matrix form,

$$\begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

so,

$$\frac{\partial}{\partial t} \vec{v}(t) = M \cdot \vec{v}(t)$$

the question now is how to solve for $\vec{v}(t)$? You may remember the derivative,

$$\frac{\partial}{\partial t}(e^{kt}) = k \cdot e^{kt}$$

In other words $f(t) = e^{kt} \cdot f(0)$ solves the equation,

$$\frac{\partial}{\partial t}(f(t)) = k \cdot f(t)$$

We should think of solving the equation,

$$\frac{\partial}{\partial t} \vec{v}(t) = M \cdot \vec{v}(t)$$

as a higher dimensional analogue of this. It turns out that,

$$\vec{v}(t) = e^{Mt} \cdot \vec{v}(0)$$

here e^{Mt} is the matrix given by plugging Mt in the *Taylor-series* for e^x ,

$$e^x = 1 + x + x^2/2 + x^3/3! + x^4/4! + \dots$$

so,

$$e^{Mt} = \text{Id} + Mt + M^2t^2/2 + M^3/3! + M^4/4! + \dots$$

After a bit of work you can show that,

$$e^{Mt} = \begin{bmatrix} 1 - t^2/2 + t^4/4! - t^6/6! + \dots & -t + t^3/3! - t^5/5! + t^7/7! - \dots \\ t - t^3/3! + t^5/5! - t^7/7! + \dots & 1 - t^2/2 + t^4/4! - t^6/6! + \dots \end{bmatrix}$$

and if you remember your Taylor series you can see that,

$$e^{Mt} = \begin{bmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{bmatrix}$$

which is the rotation matrix we have seen in class! If we have the initial condition $\vec{v}(0) = [1, 0]$ then,

$$\vec{v}(t) = e^{Mt} \cdot \vec{v}(0) = \begin{bmatrix} \cos(t) \\ \sin(t) \end{bmatrix}$$

and it turns out that as we vary t , this vector traces out the unit circle! In general idea can be used to solve any collection of equations,

$$\frac{\partial x}{\partial t} = ax(t) + by(t), \frac{\partial y}{\partial t} = cx(t) + dy(t)$$

and see that a solution is given by,

$$\vec{v}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = e^{\begin{bmatrix} a & b \\ c & d \end{bmatrix} t} \cdot \vec{v}(0)$$

If you like videos, this examples comes from, <https://www.youtube.com/watch?v=0850WBJ2ayo&t=1111s> heres another, <https://www.youtube.com/watch?v=LwSk9M5lJx4>.

8.4 Homology:

Homology is a tool used by mathematicians (and more recently by data scientists) to understand the features of a geometric space, like a graph. For example, we may make a graph of all the people in the class who know each other. We would like to understand features of this graph, like how many different groups are there where the members of each group don't know each other? This is measured by the dimension of a vector space,

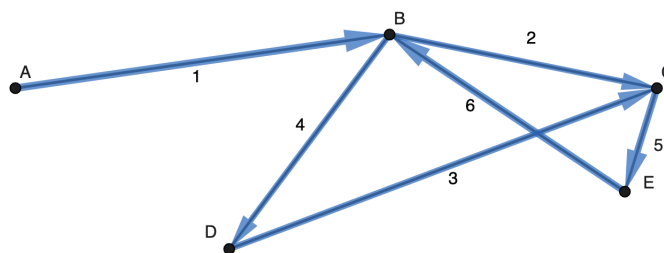
$$H_0(X)$$

How many cycles of people know each other? This is measured by the dimension of another vector space,

$$H_1(X)$$

In practice these dimensions will be huge! but in the following we show how to compute these in a (relatively) small case.

Let X be a *directed graph*, for instance we will use the following picture.



So there are vertices A, B, C, D, E and edges 1, 2, 3, 4, 5, 6 connecting them. Each edge is *oriented*, which means that we assign the values $+1, -1$ to its vertices. If the edge $x - e - y$ is directed from x to y , then the vertex x is assigned $+1$ and y is assigned -1 .

We form the *boundary matrix* ∂ as follows. There are n -columns, one for each edge e , and m -rows one for each vertex. If $x - e - y$ is the first edge, oriented from x to y , the matrix ∂ should

send,

$$\partial \cdot \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

and so on. The basis vector $\vec{b}_e$ associated to an edge is sent to the vector $\vec{b}_x - \vec{b}_y$. So in the example above,

$$\partial = ccccccc123456(cccccc)c100000A - 11010 - 1B0 - 1 - 1010C001 - 100D0000 - 11E$$

So ∂ gives a matrix,

$$\partial : \mathbb{R}^6 \rightarrow \mathbb{R}^5$$

or in general we might write,

$$\partial : \mathbb{R}^{\{\text{edges}\}} \rightarrow \mathbb{R}^{\{\text{vertices}\}}$$

we will be interested in the following subspaces,

$$\ker(\partial) = H_1(X) \subset \mathbb{R}^{\{\text{edges}\}}$$

$$\text{coker}(\partial) = H_0(X) \subset \mathbb{R}^{\{\text{vertices}\}}$$

called the **homology spaces of X** In the example above you can compute,

$$H_1(X) = \ker(\partial) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ -1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

In our picture these vectors correspond to the loops,

$$BCD \text{ and } BCE$$

notice that the -1 's correspond to going around the loop against the orientation of the edges, and the $+1$'s correspond to flowing with the orientation.

Recall that $\text{coker}(\partial) = \text{Im}(\partial)^\perp$. We can use the Rank-Nullity theorem to check that, $\dim(\text{Im}(\partial)) = 4$ which implies, $\dim(\text{coker}(\partial)) = 1$ and you can check that,

$$H_0(X) = \text{coker}(\partial) = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(which is to say the dot product of $\vec{v}$ with each of the columns is zero). So,

$$\dim(H_1(X)) = 2, \dim(H_0(X)) = 1$$

It turns out that this corresponds to the fact that our graph has **two loops** and **one connected component**.

Things are more exciting when we allow the shape X to have higher dimensional components. We might have a 2-dimensional faces connecting the edges, and a 3-dimensional volume filling in the faces. Imagine a tetrahedron, or a bunch of triangles randomly glued together in 3-dimensions. Spaces like these are called **simplicial complexes**. For spaces like these we can associate subspaces,

$$H_n(X) \subset R^{\#n\text{-dim simplices}}$$

whose dimension counts the number of n -dimensional "holes" in the spaces.

This looks like a good introduction, <https://www.math.uchicago.edu/~may/VIGRE/VIGRE2007/REUPapers/FINALFULL/Nadathur.pdf> or if you like videos <https://www.youtube.com/watch?v=5xLe77iTHuQ&t=579s> but like all of these topics you should look around!

8.5 Markov Chains:

Markov chains are used throughout science, probability theory and dynamical systems. The following is an example motivated by biology from the new MATH 16 course at UCSC. See the "Two-Compartment Markov Chain Explorer" at <https://martyweissman.github.io/>.

Suppose we are interested in how squirrels move around in the forest. In total there are 1000 squirrels, and we divide the forest into two sections A, B . Every day two things happen,

1. 25% of the squirrels in section A go to B , the rest stay in A
2. 40% of the squirrels in section B go to A , the rest stay in B .

If we write $A = \#\{\text{squirrels in } A\}$, $B = \#\{\text{squirrels in } B\}$ we can express the above as the following linear maps,

$$A \mapsto 0.75A + 0.4B$$

$$B \mapsto 0.25A + 0.6B$$

which can be written in matrix form,

$$M = \begin{bmatrix} 0.75 & 0.4 \\ 0.25 & 0.6 \end{bmatrix} \cdot \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0.75A + 0.4B \\ 0.25A + 0.6B \end{bmatrix}$$

So after k days, the resulting population of squirrels is,

$$M^k \cdot \begin{bmatrix} A \\ B \end{bmatrix}$$

We will be interested in what the long term behavior of M is. In other words, what happens in the limit?

$$\lim_{k \rightarrow \infty} M^k = ?$$

Now, the matrix M turns out to be **diagonalizable**. Recall this means there is a change of basis

matrix $N = \begin{bmatrix} 8 & 1 \\ 5 & -1 \end{bmatrix}$ such that,

$$NMN^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 7/20 \end{bmatrix}$$

Where 1 and 7/20 are the eigenvalues of M . In any case, one can check that,

$$(NMN^{-1})^k = NM^kN^{-1} = \begin{bmatrix} 1^k & 0 \\ 0 & (7/20)^k \end{bmatrix}$$

So in the limit we get,

$$\lim_{k \rightarrow \infty} NM^kN^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

So,

$$\lim_{k \rightarrow \infty} M^k = N^{-1} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot N = \frac{1}{13} \begin{bmatrix} 8 & 8 \\ 5 & 5 \end{bmatrix}$$

and you can see that,

$$\text{Image}(\lim_{k \rightarrow \infty} M^k) = \text{span}\left\{ \begin{bmatrix} 8 \\ 5 \end{bmatrix} \right\}$$

In words, for any starting state $[A, B]$, over time the number of squirrels in each section of the forest tends towards a ratio of $8A = 5B$. If we started with 1000 total squirrels, then in the limit we should see that,

$$A + B = 1000$$

$$8A = 5B$$

Solving these equations gives,

$$A = 5000/13 \approx 384 \text{ squirrels in section } A$$

$$B = 8000/13 \approx 616 \text{ squirrels in section } B$$

In vague terms, a (discrete) Markov chain, is a system with several compartments $A_1, \dots, A_n$ and

at each time step, there is some probability for things to move between compartments. If we write

$$P(1 \rightarrow 5)$$

for the probability of moving from compartment 1 to 5 etc. Arranging the probabilities $P(i \rightarrow j)$ into a matrix gives a system similar to what we have seen above,

$$M = \left[P(i \rightarrow j) \right]_{i,j=1,\dots,n}$$

and we can try to study its long term behavior by studying what happens in the limit,

$$\lim_{k \rightarrow \infty} M^k$$

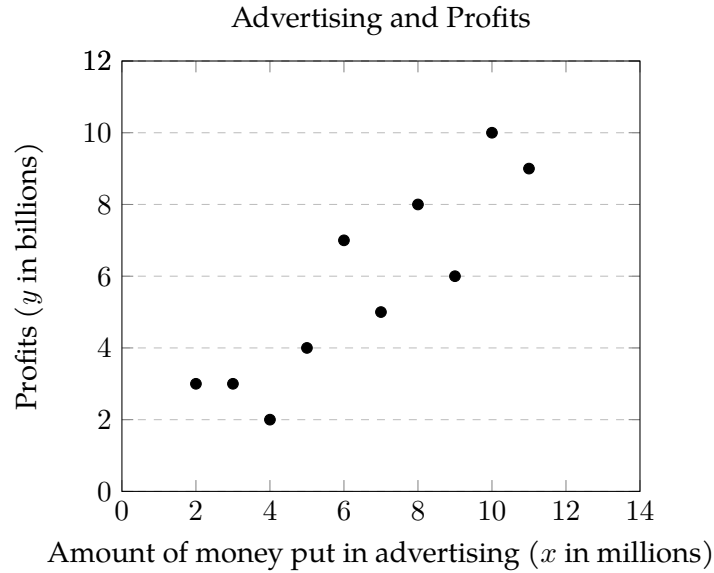
8.6 Linear Regression:

Linear Regression is a powerful way in data science for studying how the input factors affect the outcomes. It has many applications in Business Management, Economy, etc. For a quick preview to get the idea, see the video "How Linear Regression Works" at <https://youtu.be/qUfVrYKetfA?si=SCeOPHFShZgwuzV>.

Suppose that we are interested in the problem of how the amount of money put into advertising would affect the profits of a company that sells frozen pizza. The first thing we do is to collect several data, say the data are given by

Advertising and Profits	
Amount of money put in advertising (millions)	Profits (billions)
2	3
3	3
4	2
5	4
6	7
7	5
8	8
9	6
10	10
11	9

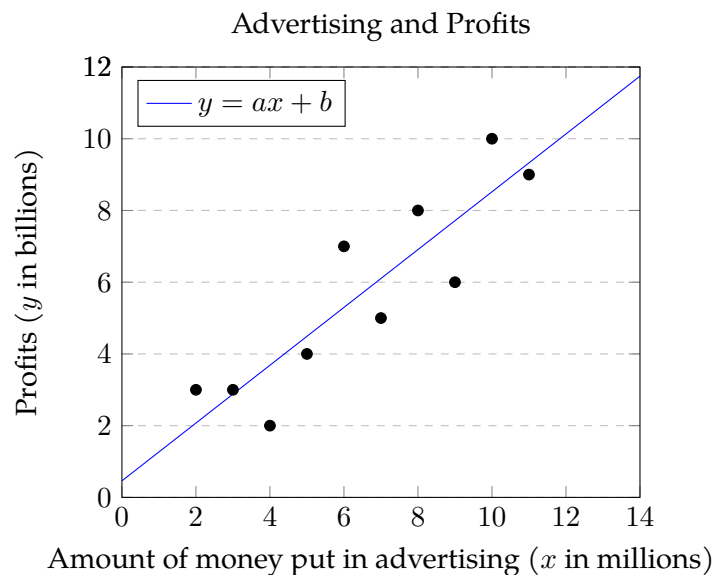
We plot them with each pair (x, y) representing a data.



From the plots, it looks like there's a linear relationship between x and y . This is a very important assumption for doing linear regression; we do linear regression only when the data suggest a linear relationship between the variables. Back to our example, it is obvious that there is no way to find out a linear equation that passes through all the points. The best we can do is to find out a linear equation that "best fit" all the points. (Note: If the data shows other relationship, for example " $y = x^2$ ", we would have to find a parabola that best fit the data)

For convenience, let's denote the data points by $(x_1, y_1), (x_2, y_2), \dots, (x_{10}, y_{10})$. The goal is to find a straight line $y = ax + b$ (called **the least square straight line**) so that the sum of the "distance square" of each point to the straight line is the smallest, that is, we want to find a and b so that the following is the smallest:

$$E = \sum_{i=1}^{10} [y_i - (ax_i + b)]^2$$



This is called **the least squares problem** for $(x_1, y_1), (x_2, y_2), \dots, (x_{10}, y_{10})$, and using the language of matrices and column vectors, we can solve the least squares problem in a very elegant way.

(a) Let $\vec{y_0} = \begin{bmatrix} 3 \\ 3 \\ 2 \\ 4 \\ 7 \\ 5 \\ 8 \\ 6 \\ 10 \\ 9 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \\ 6 & 1 \\ 7 & 1 \\ 8 & 1 \\ 9 & 1 \\ 10 & 1 \\ 11 & 1 \end{bmatrix}$, and $\vec{v} = \begin{bmatrix} a \\ b \end{bmatrix}$.

That is, $\vec{y_0}$ consists of the y_i values and the first column of A consists of the x_i values.

We can simply verify that

$$E = \|\vec{y}_0 - A\vec{v}\|^2$$

and so the least squares problem becomes finding $\vec{v}_0 \in \mathbb{R}^2$ such that $\|\vec{y}_0 - A\vec{v}_0\| \leq \|\vec{y}_0 - A\vec{v}\|$ for any other $\vec{v} \in \mathbb{R}^2$.

- (b) Let W be the range of A , and let $\vec{u}_0$ be the projection of $\vec{y}_0$ onto W . Then, $\|\vec{y}_0 - \vec{u}_0\|$ is the closet distance between $\vec{y}_0$ and W . Since $\vec{u}_0 \in W$, we can find out an $\vec{v}_0 \in \mathbb{R}^2$ such that $A\vec{v}_0 = \vec{u}_0$. By the fact that $\|\vec{y}_0 - \vec{u}_0\|$ is the closet distance between $\vec{y}_0$ and W , we know that $\|\vec{y} - A\vec{v}_0\| \leq \|\vec{y} - A\vec{v}\|$ for any other $\vec{v} \in \mathbb{R}^2$. This $\vec{v}_0$ is then our solution to the least squares problem.

- (c) To practically calculate $\vec{v}_0$, one first show that $A^T(A\vec{v}_0 - \vec{y}_0) = \vec{0}$ by the general property

$$(A\vec{x}) \cdot \vec{y} = \vec{x} \cdot (A^T\vec{y})$$

for any vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ and any $m \times n$ matrix A , together with the property

$$\left\{ \vec{x} \in \mathbb{R}^n \mid \vec{y} \cdot \vec{x} = \vec{0} \text{ for all other } \vec{y} \in \mathbb{R}^n \right\} = \left\{ \vec{0} \right\}$$

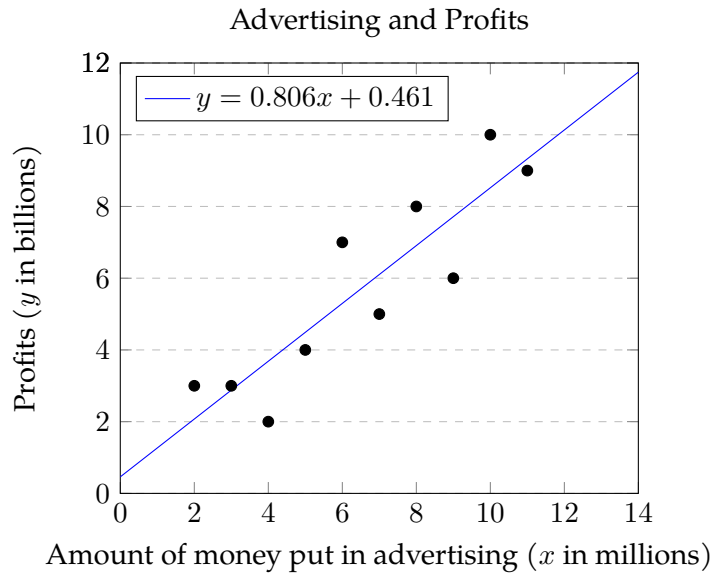
Therefore, by the fact that $A^T(A\vec{v}_0 - \vec{y}_0) = \vec{0}$, we know $\vec{v}_0$ satisfies $A^T A\vec{v}_0 = A^T \vec{y}_0$. This means $\vec{v}_0 = (A^T A)^{-1} A^T \vec{y}_0$.

(Note: $A^T A$ is always invertible while doing linear regression; this can be proven by calculating the rank of A with the fact that usually the first column of A is not constant, and then using the property $\text{rank}(A^T A) = \text{rank}(A)$)

- (d) Back to our example, we can calculate the solution $\vec{v}_0$ by using the formula in (c). The result is

$$\vec{v}_0 = \begin{bmatrix} 0.806 \\ 0.461 \end{bmatrix}$$

This means the straight line $y = 0.806x + 0.461$ is the straight line that's best fit the data points, and it shows how the profit will increase as the amount of money put in advertising increases.



- (e) Now that we get the least square straight line, another thing we can do is prediction. For example, if we put 12 millions in advertising, the profit in billions we would get is approximately

$$0.806(12) + 0.461 = 10.133$$

Note that this way of prediction only works for x values that are close to our x_i values because the least square straight line is derived from the data $(x_1, y_1), (x_2, y_2), \dots, (x_{10}, y_{10})$, and it is the one that best fit only for the given data $(x_1, y_1), (x_2, y_2), \dots, (x_{10}, y_{10})$.